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Chapter 1

Model Builder

1.1 Instantiation Dialog

This module contains the implementation of the gui logic for the Instantiation dialog.

class **`InstantiationDlg`** (*parent=None, var_data_all=[], entity=None, variables=None, single_node=True*)

Implementation of the logic for the insantiation dialog.

`ui` Contains the GUI widgets, obtained from QtDesigner.

Type object

`entity`

Stores the information about the entity to be instantiated.

Type dict

`variables`

Stores the information about all variables in the ontology.

Type dict

`var_data`

Contains one list for variables that need to be initialized and another for the rest. Each of these lists contain tuples of two items:

- var_info (dict): Data of a single variable with the following fields.
  - id (int): The variable id.
  - same_in_all_nodes (bool): True if the variable has been instantiated with the same value in all nodes or has not been instantiated in any. False if it has been instantiated in all nodes but the value is different in at least one of them.
  - value (float or None): The value of the variable if it has been instantiated with the same value in all nodes. None otherwise.
- is_enabled (bool): If True, the user will be able to interact with the item. If False, interactions are not allowed.

Type list[list[tuples]]

`lists`

Contains the QListWidgets in the interface.

Type list[QListWidget]

tab_index_selected
Stores the index of the active tab.

Type int

__init__ (*parent=None, var_data_all=[], entity=None, variables=None, single_node=True*)
Initializes the InstantiationDlg with the required information.
Parameters

_add_new_item_to_list (*list, item_name, is_enabled*)
Adds a new item to a QListWidget.
Uses flags to control some of the item properties.

Parameters
- **list** (*QListWidget*) – Widget where the item will be inserted.
- **item_name** (*string*) – Name of the item to be inserted. Already includes the variable name and its value.
- **is_enabled** (*bool*) – If True, the user will be able to interact with the item. If False, interactions are not allowed.

_instantiate ( )
Controls the instantiation process.
Saves the new value for the selected variable. If the variable instantiated is not one of the required ones a new analisis is performed to check what variables are still needed and can still be instantiated.

_set_instantiation_state (*is_allowed*)
Controls the use of GUI widgets related to the instantiation.

Parameters **is_allowed** (*bool*) – If True the name of the selected variable will be displayed, the QTextEdit will accept a value and the Instantiate button can be pressed. If False the variable label will show a "?" sign and no interaction is allowed with the QtextEdit or the Instantiate button.

_tab_changed (*current_tab_index*)
Controls the tab change.

Parameters **tab_index** (*int*) – Index of the active tab.

_update_enabled_vars ( )
Checks what variables are still needed.
A new VarEq tree is generated. Variables that does not required instantiation but still get a value will not need equations anymore. These equations are added to the entity's original blocked equations and the tree is constructed from there.

_update_list ( )
Updates the QListWidgets with the current info.

_update_tab_title ( )
Update the first tab title when needed.
The title will be in the form: "Required Variables (#)" where # is an integer representing how many variables are still required to be instantiated.

_variable_selected (*item_pos*)
Prepares the GUI for the instantiation when a variable is selected.

Parameters **item_pos** (*int*) – Position in the QListWidget of the selected variable. It corresponds to the index in the proper var_data list.

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